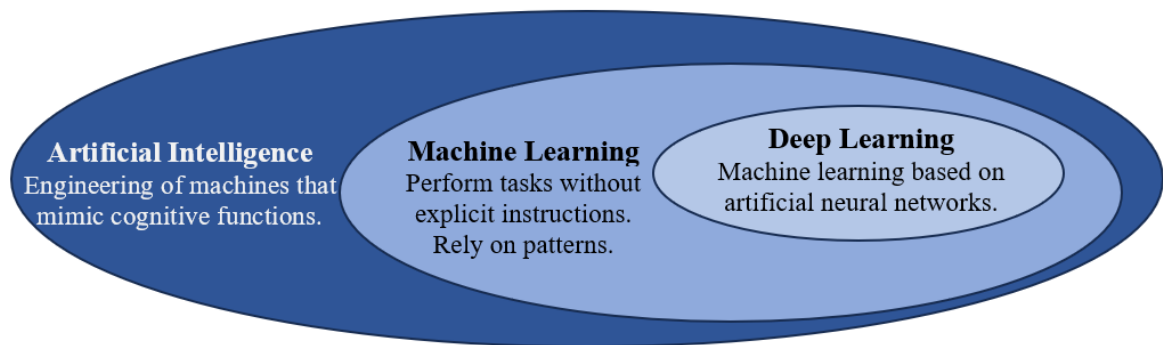


Artificial intelligence, is a field of computer science that aims to make machines that can perform tasks that require human intelligence. Machine learning is a subset of AI focusing on creating statistical models and algorithms that let computers learn from data. Deep learning is a subset of ML inspired by the human brain's structure and function. It focuses on using artificial neural networks to model and solve complex tasks.



Comparison Table of AI Vs ML Vs DL

Feature	Artificial Intelligence	Machine Learning	Deep Learning
Approach and Learning	Broad field of creating intelligent machines	AI technique that learns from data	ML technique using deep neural networks
Dependency on Data	Not always	Yes	Yes
Feature Extraction	Rule-based or manual	Manual	Automatic
Data Requirement	Low to moderate	Moderate	Large
Best For	Reasoning, planning, automation	Prediction and classification	Images, speech, natural language, and complex pattern recognition

Machine Learning Example: Email Spam Detection

Example

Suppose we receive an email with following text:

Subject: *Congratulations! You've won a free iPhone. Click here to claim your prize.*

The machine learning model detects features such as promotional language ("Congratulations!", "free", "claim your prize"), suspicious links, and an unknown sender. Based on patterns learned from previously labeled emails, it classifies the message as spam and moves it to the Spam folder instead of your inbox.

This application demonstrates how machine learning helps protect millions of users every day by automatically filtering unwanted and potentially harmful emails.

How It Works

The machine learning model is trained on thousands of emails labeled as spam or not spam with the help of different algorithms like Logistic Regression or Decision Tree. It learns patterns based on features such as:

- Words and phrases commonly found in spam emails (e.g., "Congratulations!", "Win now!", "Limited-time offer")
- Sender reputation and email domain
- Number of links and attachments
- Email formatting and unusual characters
- User behavior, such as marking emails as spam or moving them to the inbox

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Why Machine Learning is Suitable in Email Spam Detection?	Why Deep Learning is an overkill and unnecessary in Email Spam Detection?
Email data is largely structured into measurable features.	DL solutions are effective in complex scenarios involving large datasets. The solution will be an overkill for simple email spam detection task and so unnecessary.
The important characteristics can be manually extracted.	Training is computationally expensive.
Training requires relatively less computational power.	The improvement in accuracy is often small compared to traditional ML models.
The model is easy to update as new spam patterns emerge.	The model is more difficult to interpret.
It provides interpretable results, helping developers understand why an email was flagged.	

Deep Learning Example: Self-Driving Car Object Detection

Example

Imagine a self-driving car approaching an intersection. The deep learning system simultaneously identifies a red traffic light, a pedestrian crossing the road, and a cyclist approaching from the right. Based on this information, it slows down, stops at the red light, waits for the pedestrian to cross safely, and then proceeds when the light turns green and the path is clear. This ability to interpret complex visual scenes and make safe driving decisions in real time demonstrates why deep learning is a critical technology for autonomous vehicles.

One of the most important real-world applications of deep learning is autonomous (self-driving) vehicles. Companies such as Tesla, Waymo, Mercedes-Benz, and NVIDIA use deep learning to help vehicles perceive their surroundings, understand road conditions, and make driving decisions with minimal or no human intervention.

How It Works

Self-driving cars are equipped with cameras, radar, LiDAR, GPS, and ultrasonic sensors that continuously collect data from the vehicle's surroundings. Deep learning models—particularly **Convolutional Neural Networks (CNNs)** for image recognition and other neural network architectures for perception and decision-making analyze this data in real time to:

- Detect pedestrians, cyclists, and other vehicles.
- Recognize traffic lights and road signs.
- Identify lane markings and road boundaries.
- Estimate the speed and distance of nearby objects.
- Predict the movement of surrounding vehicles and pedestrians.
- Decide when to accelerate, brake, or steer.

The system learns these tasks by training on millions of images, videos, and driving scenarios, allowing it to recognize complex patterns that are difficult to program using explicit rules.

Why Deep Learning Is Suitable?	Why Traditional Machine Learning Is Less Suitable?
Images and video are highly unstructured data.	It requires manually designing image features (such as edges, corners, or textures).
Manual feature extraction is extremely difficult.	Handcrafted features often fail under varying real-world conditions.
CNNs automatically learn complex visual patterns.	Performance decreases significantly with complex scenes.
Deep learning achieves very high accuracy in image recognition tasks.	It struggles to process large volumes of high-dimensional image data.
It can handle changing lighting, weather, viewing angles, and partially obscured objects.	